# Response to the Reviewers

Paper ID 1343: *Real-Time Adaptive Cross-Media Recommendation Using Semantic Embeddings, Knowledge Graph Scoring, and Graph Collaborative Learning*

International Conference on Machine Learning and Data Engineering (ICMLDE 5.0)

Dear Scientific Committee and Reviewers,

We thank the reviewers for their careful reading and constructive comments. We have revised the manuscript and respond to every comment below, quoting each comment in italics and giving what we changed and where. The main changes are:

- **Larger evaluation** (Section 4 (p. 6), Section 5 (p. 7)): three Amazon Reviews 2023 datasets with up to 50,000 users, reporting Recall, NDCG, MAP, and MRR at $K \in \{10, 20\}$ (plus HitRate and Precision) with significance tests (Table 2 (p. 7)), and 17 prototype scenarios instead of 5 (Table 6 (p. 9)).
- **Full ablation**: all 16 combinations of the four signals (semantic, EMA, LightGCN, knowledge graph), each with weights selected on a validation split (Table 4 (p. 8)).
- **Baselines**: most-popular and single-signal rankings evaluated under the same protocol as the hybrid (Table 4 (p. 8)).
- **Configuration and latency**: complete experimental settings (Table 3 (p. 7)) and measured latency and throughput (Table 5 (p. 9)).
- **Method descriptions**: event weights, EMA initialization and negative updates, LightGCN training, and the knowledge-graph construction (Section 3 (p. 3)).
- **Related work**: Table 1 (p. 3) updated and Section 2 (p. 2) extended with 2025–2026 work.
- **Format and language**: Procedia Computer Science template, redrawn figures, shortened captions, and proofreading of the entire manuscript.

# Numbering in the submitted and revised manuscripts

Section, table, figure, and equation numbers differ between the two versions, so each comment below lists its location in both. Page numbers of the submitted version are those of the PDF we submitted; page numbers of the revised version are those of the attached manuscript. A table or figure that appears under the same number in both versions may show different content, as noted.

| | Submitted version | Revised version |
|---|---|---|
| Introduction | Section 1 (p. 2) | Section 1 (p. 1) |
| Related work | Section 2, "Discussion of Related Work and Novelty" (pp. 2–3) | Section 2, "Related Work and Novelty" (p. 2) |
| System overview | Sections 3.1 and 3.9 (pp. 4, 8) | Section 3.1 (p. 3) |
| Data and catalog | Section 3.2 (p. 4) | Section 3.2 (p. 3) |
| Semantic retrieval, creator and title matching | Sections 3.3 and 3.4 (p. 5) | Section 3.3 (p. 4) |
| EMA | Section 3.5 (p. 6) | Section 3.4 (p. 4) |
| LightGCN | Section 3.6 (p. 6) | Section 3.5 (p. 4) |
| Knowledge graph | Section 3.7 (p. 7) | Section 3.6 (p. 5) |
| Hybrid ranking | Section 3.8 (p. 8) | Section 3.7 (p. 6) |
| Experimental setup | — (new section) | Section 4 (p. 6) |
| Results | Section 4 (p. 9) | Section 5: baselines and ablation 5.1, latency 5.2, prototype scenarios 5.3 (p. 7) |
| Conclusion | Section 5 (p. 10) | Section 6 (p. 10) |
| Table 1 | Comparison of prior model families and measures (p. 3) | Table 1, *updated*: comparison with recent graph, knowledge-aware, and cross-domain recommenders (p. 3) |
| Table 2 | Module-wise input, output, and algorithm (p. 8) | Removed; the content is in Section 3 and Fig. 1 |
| Table 2 (revised) | — | Table 2, benchmark graphs (p. 7) |
| Table 3 | Prototype results with qualitative single-signal rows (p. 9) | Qualitative rows replaced by the quantitative Table 4 (p. 8); per-scenario results in Table 6 (p. 9) |
| Table 3 (revised) | — | LightGCN and EMA settings (p. 7) |

| | Submitted version | Revised version |
|---|---|---|
| Table 5 (revised) | — | Latency (p. 9) |
| Figure 1 | Architecture (p. 4) | Fig. 1, redrawn (p. 3) |
| Figure 2 | Knowledge graph (p. 7) | Fig. 2, redrawn (p. 5) |
| Figure 3 | Result bar chart (p. 9) | Removed; its values are in Table 6 |
| Equations | (1) catalog, (2) semantic score, (3) EMA update, (4) EMA score, (5)–(6) LightGCN, (7) BPR, (8) graph score, (9) paths, (10) KG score, (11) profile score, (12) hybrid | (1) catalog, (2) EMA update, (3) LightGCN, (4) BPR, (5) paths, (6) KG score, (7) profile score, (8) hybrid, (9) preference bonus (new); the semantic, EMA, and graph scores are now defined in the text |

# Reviewer 1

**Comment R1.1.** *"The Figure 1 caption is written twice."*

**Response.** We redrew Figure 1 without embedded text, so it now has a single caption, and updated it to the current data sources.

**Submitted version:** Fig. 1, p. 4.

**Revised version:** Fig. 1 (p. 3).

**Comment R1.2.** *"Different text fonts are used on pages 4 and 5."*

**Response.** We changed the font of the entire manuscript to Times, the template font, including the text inside both figures.

**Submitted version:** pp. 4–5 (Sections 3.1–3.4 and Fig. 1); the same typeface also appeared on later pages.

**Revised version:** Entire manuscript; Section 3 (pp. 3–6), Fig. 1 (p. 3), Fig. 2 (p. 5).

**Comment R1.3.** *"Check the caption of Figure 2."*

**Response.** We shortened the caption to one sentence and redrew Figure 2 so that its relation labels match the relations defined in the knowledge-graph subsection.

**Submitted version:** Fig. 2, p. 7 (Section 3.7).

**Revised version:** Fig. 2 (p. 5), Section 3.6 (p. 5).

**Comment R1.4.** *"Correct the captions of all tables."*

**Response.** All table captions now follow the template: one line above the table, with explanatory text moved to the body. Tables 1–3 of the submitted version correspond to revised Tables 1, 4, and 6 (the submitted Table 2 was removed); the new tables follow the same rule.

**Submitted version:** Tables 1–3 (pp. 3, 8, 9).

**Revised version:** Tables 1–6 (pp. 3–9).

**Comment R1.5.** *"Cite papers from 2025 onwards."*

**Response.** We added six 2025–2026 papers: LGKAT [18], LightKG [19], LLMCDSR [20], LLM4CDSR [21], LeCDSR [22], and KLLMRec [23]. They are discussed in Section 2 and compared in Table 1; the works [11, 16, 17] are also from 2025–2026.

**Submitted version:** Section 2 (pp. 2–3); References [1]–[20].

**Revised version:** Section 2 (p. 2); Table 1 (p. 3); References [18]–[23].

**Comment R1.6.** *"Spelling mistakes (e.g., in conclusion)."*

**Response.** We corrected the following five mistakes and proofread the entire manuscript:

(1) "papaer" → "paper" (Conclusion);

(2) "differnent" → "different" (Section 2);

(3) "retrival" → "retrieval" (Section 2);

(4) "choosen" → "chosen" (Section 2);

(5) "go show" → "to show" (Introduction).

The citation that printed as "[?]" in Section 2 (the SASRec reference) was also fixed.

**Submitted version:** "papaer": Section 5, p. 10; "differnent", "choosen", and "[?]": Section 2, p. 3; "retrival": Section 2, p. 2; "go show": Section 1, p. 2.

**Revised version:** Section 6 (p. 10); Section 2 (p. 2); Section 1 (p. 1); entire manuscript.

**Comment R1.7.** *"Ensure the paper is in the template."*

**Response.** The manuscript now uses the Procedia Computer Science template without modifications: custom spacing, font, caption, and list settings were removed, and the running head, the corresponding-author footnote, and the reference style follow the template.

**Submitted version:** Entire manuscript.

**Revised version:** Entire manuscript.

# Reviewer 2

**Comment R2.1.** *"The proposed integration of Sentence-BERT, Qdrant, EMA personalization, LightGCN, and knowledge-graph/profile scoring is interesting and clearly presented. However, the manuscript should strengthen the novelty discussion by providing a more detailed comparison with recent cross-domain and hybrid recommendation approaches."*

**Response.** We updated Table 1 (it now compares LightGCN, KGAT, LGKAT, LightKG, LLMCDSR, LLM4CDSR, and this work on domains, text semantics, collaborative graph, knowledge graph, short-term adaptation, and deployment) and extended Section 2 with recent knowledge-aware graph recommenders [18, 19] and LLM-based cross-domain recommenders [20–23]. The novelty paragraph now states that the contribution is the integration of these signals over one catalog, with separate immediate (EMA) and periodic (LightGCN) personalization, together with the trade-offs of this design.

**Submitted version:** Section 2 (pp. 2–3); Table 1 (p. 3).

**Revised version:** Section 2 (p. 2); Table 1 (p. 3).

**Comment R2.2.** *"The experimental evaluation is limited to five prototype scenarios. Please provide a larger-scale evaluation using standard recommendation metrics such as Recall@K, NDCG@K, MAP, and MRR, as the current study itself identifies these as future evaluation requirements."*

**Response.** We extended the evaluation in two ways.

(1) *Offline benchmark.* Three Amazon Reviews 2023 datasets [27]: Movies & TV (50,000 users, 52,213 items), Health & Household (50,000 users, 42,621 items), and Industrial & Scientific (33,959 users, 18,804 items). Each method ranks the full catalog for every test user. We report Recall, NDCG, MAP, and MRR at $K = 10$ and $K = 20$, together with HitRate and Precision, and paired Wilcoxon signed-rank tests.

(2) *Prototype scenarios.* We increased the scenarios from 5 to 17 (five person queries, six cross-media intents, and six queries in the health, industrial, and finance verticals), ran all of them through the deployed API, and added Precision, Recall, NDCG, MAP, and MRR at 10 to the earlier measures. The results of every scenario are in Table A below.

**Submitted version:** Section 4 (p. 9); Table 3 (p. 9); Fig. 3 (p. 9).

**Revised version:** Section 4 (p. 6), Table 2 (p. 7); Section 5.1 (p. 7), Table 4 (p. 8); Section 5.3 (p. 9), Table 6 (p. 9).

**Table A.** Results of the 17 prototype scenarios (PH: PersonHit, PA: ProfileAlign, KG and Prof.: mean KG and profile score of the top five; P, R, N, MAP, MRR: rank metrics at 10 against rule-based relevance judgments, defined in Section 5.3 (p. 9)).

| Scenario | Age / domain | PH@5 | PA@5 | KG | Prof. | P@10 | R@10 | N@10 | MAP@10 | MRR@10 | Content types in top 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| James Cameron | 20 | 4/5 | 5/5 | 0.667 | 0.980 | 0.800 | 0.533 | 0.716 | 0.617 | 0.500 | finance, movie, show |
| Taylor Swift | 20 | 5/5 | 5/5 | 0.667 | 0.630 | 0.800 | 0.216 | 0.870 | 0.800 | 1.000 | book, movie, music |
| Business analytics | 30 | – | 5/5 | 0.650 | 1.000 | 0.900 | 0.016 | 0.927 | 0.866 | 1.000 | book, finance |
| Fun animation adventure | 8 | – | 5/5 | 0.667 | 1.000 | 1.000 | 0.002 | 1.000 | 1.000 | 1.000 | movie |
| Space adventure with aliens | 20 | – | 5/5 | 0.600 | 0.896 | 1.000 | 0.006 | 1.000 | 1.000 | 1.000 | movie, show |
| Joint pain supplements | 30, health | – | 5/5 | 0.900 | 1.000 | 1.000 | 0.003 | 1.000 | 1.000 | 1.000 | health |
| Lab safety gloves | 30, industry | – | 5/5 | 0.767 | 0.694 | 1.000 | 0.010 | 1.000 | 1.000 | 1.000 | industrial |
| Personal budgeting | 30, finance | – | 5/5 | 0.700 | 0.938 | 1.000 | 0.011 | 1.000 | 1.000 | 1.000 | finance |
| Christopher Nolan | 30 | 5/5 | 5/5 | 0.667 | 0.640 | 0.800 | 0.533 | 0.848 | 0.733 | 1.000 | movie |
| Stephen King | 30 | 4/5 | 5/5 | 0.667 | 0.834 | 0.900 | 0.125 | 0.890 | 0.790 | 1.000 | book, show |
| Coldplay | 20 | 4/5 | 5/5 | 0.333 | 0.502 | 0.900 | 0.237 | 0.780 | 0.707 | 0.500 | movie, music |
| Magic and dragons | 20 | – | 5/5 | 1.000 | 0.958 | 1.000 | 0.007 | 1.000 | 1.000 | 1.000 | book, movie, show |
| Romantic drama about love | 30 | – | 5/5 | 0.550 | 0.992 | 1.000 | 0.002 | 1.000 | 1.000 | 1.000 | movie, show |
| Action with superheroes | 30 | – | 5/5 | 0.650 | 0.970 | 1.000 | 0.003 | 1.000 | 1.000 | 1.000 | movie |
| Vitamin D and immunity | 30, health | – | 5/5 | 1.000 | 0.938 | 1.000 | 0.003 | 1.000 | 1.000 | 1.000 | health |
| Torque wrench and fasteners | 30, industry | – | 5/5 | 0.760 | 0.854 | 1.000 | 0.012 | 1.000 | 1.000 | 1.000 | industrial |
| Retirement investing | 30, finance | – | 5/5 | 0.600 | 0.758 | 1.000 | 0.014 | 1.000 | 1.000 | 1.000 | finance |
| Mean | | 0.88 | 1.00 | 0.697 | 0.858 | 0.947 | 0.102 | 0.943 | 0.913 | 0.941 | 8/17 cross-domain |

**Comment R2.3.** *"A detailed ablation study is recommended to quantify the individual contribution of semantic retrieval, EMA personalization, LightGCN, and KG/profile scoring. This would help demonstrate whether the improvement is actually attributable to the proposed hybrid architecture. The final score currently combines these components using fixed weights."*

**Response.** We added a full ablation of the four signals (semantic S, EMA E, LightGCN G, knowledge graph K): all 16 on/off combinations, evaluated on the three datasets with Recall, NDCG, MAP, and MRR (Table 4). The empty combination is most-popular ranking and S+E+G+K is the proposed hybrid. The weights are no longer fixed: each combination has its own weights, selected by grid search on a separate validation split, and the hand-set weights are listed as a reference row. The declared-profile score has no counterpart in public interaction logs, so it is ablated in the prototype scenarios (see R3.6). Table B gives NDCG@20 and Recall@20 of the 16 combinations: on Movies & TV and Industrial & Scientific the four-signal hybrid is the best of all 16 on every measure, and on Health & Household the knowledge-graph signal alone is best.

**Submitted version:** Section 3.8, Eq. (12) (p. 8); Section 4 (p. 9).

**Revised version:** Section 4 (p. 6) (baselines and weight selection); Section 5.1 (p. 7); Table 4 (p. 8); Section 5.3 (p. 9).

**Table B.** NDCG@20 (N) and Recall@20 (R) of all 16 combinations of the four signals; the best value in each column is in bold, and ° marks a combination whose NDCG@20 does not differ significantly from S+E+G+K ($p \geq 0.05$). The full table with MAP and MRR is Table 4.

| Signals | Movies & TV | | Health & Household | | Industrial & Scientific | |
|---|---|---|---|---|---|---|
| | R@20 | N@20 | R@20 | N@20 | R@20 | N@20 |
| None (popularity) | 0.0197 | 0.0097 | 0.0237 | 0.0115 | 0.0282 | 0.0147 |
| S | 0.0307 | 0.0187 | 0.0166 | 0.0096 | 0.0379 | 0.0223 |
| E | 0.0233 | 0.0128 | 0.0104 | 0.0052 | 0.0254 | 0.0135 |
| G | 0.0257 | 0.0136 | 0.0116 | 0.0054 | 0.0242 | 0.0130 |
| K | 0.0093 | 0.0045 | **0.0253** | **0.0144** | 0.0364 | 0.0210 |
| S+E | 0.0309 | 0.0188 | 0.0169 | 0.0097 | 0.0380 | 0.0225 |
| S+G | 0.0359 | 0.0222 | 0.0212 | 0.0120 | 0.0468 | 0.0267 |
| S+K | 0.0313 | 0.0193 | 0.0170 | 0.0097 | 0.0387 | 0.0226 |
| E+G | 0.0307 | 0.0174 | 0.0187 | 0.0092 | 0.0377 | 0.0205 |
| E+K | 0.0247 | 0.0141 | 0.0144 | 0.0072 | 0.0314 | 0.0162 |
| G+K | 0.0265 | 0.0141 | 0.0200 | 0.0097 | 0.0386 | 0.0207 |
| S+E+G | 0.0362 | 0.0224 | 0.0215 | 0.0122 | 0.0490 | 0.0277 |
| S+E+K | 0.0318 | 0.0197 | 0.0171 | 0.0098 | 0.0389 | 0.0227 |
| S+G+K | 0.0366 | 0.0226 | 0.0224 | 0.0126° | 0.0484 | 0.0275 |
| E+G+K | 0.0314 | 0.0177 | 0.0232 | 0.0115 | 0.0410 | 0.0225 |
| S+E+G+K | **0.0368** | **0.0227** | 0.0223 | 0.0126 | **0.0495** | **0.0283** |
| Hand-set weights | 0.0344 | 0.0216 | 0.0208 | 0.0122 | 0.0455 | 0.0266 |

**Comment R2.4.** *"The baseline comparison should be strengthened. Table 3 mainly provides qualitative descriptions of semantic-only, graph-only, and KG/profile-only systems. Implementing these baselines under the same experimental conditions and reporting comparable quantitative results would provide stronger evidence for the effectiveness of the proposed approach."*

**Response.** *Quantitative:* the most-popular, semantic-only, EMA-only, LightGCN-only, and KG-only baselines are implemented in the same evaluation code as the hybrid, with the same chronological split, the same full-catalog candidate

set, and the same metrics; their results are the first rows of Table 4 and are compared with the hybrid in Section 5.1 (p. 7). *Qualitative:* Section 4 now describes what each baseline captures and what it misses, and Section 5.1 discusses how each behaves on each dataset. The qualitative rows of the submitted Table 3 were removed.

**Submitted version:** Table 3 (p. 9).

**Revised version:** Section 4 (p. 6) (baselines); Section 5.1 (p. 7); Table 4 (p. 8).

**Comment R2.5.** *"The reproducibility and real-time claims should be strengthened. Please provide complete experimental configurations, including data splitting, LightGCN training parameters, EMA settings, hybrid-score weight selection, and detailed latency/throughput measurements. The manuscript currently acknowledges that production latency benchmarking and larger-scale evaluation remain future work."*

**Response.** Section 4 now specifies the data preparation (rating threshold, $k$-core filtering, user sampling), the chronological split (last two positives for testing, the previous one for validation), the candidate set, the offline counterpart of each signal, the metrics, and the weight-selection grid. Table 3 lists all LightGCN and EMA settings and the training time. Section 5.2 and Table 5 report latency and throughput measured on 700 requests to the deployed endpoint over the full 120,747-item catalog: median 363 ms (p95 455 ms, p99 507 ms), EMA update 0.05 ms, and 2.7–3.3 requests per second per process. The conclusion no longer lists these evaluations as future work.

**Submitted version:** Section 4 (p. 9); Section 5 (p. 10).

**Revised version:** Section 4 (p. 6); Table 2 (p. 7); Table 3 (p. 7); Section 5.2 (p. 8); Table 5 (p. 9); Section 6 (p. 10).

# Reviewer 3

**Comment R3.1.** *"Specify LightGCN's training settings, including layers, embedding size, negative sampling, and stopping criteria."*

**Response.** The settings are now stated in the text and in Table 3: 64-dimensional embeddings, 3 propagation layers (averaged), one uniformly sampled negative per positive pair (resampled if observed), Adam with learning rate $10^{-3}$ and $L_2$ weight $10^{-4}$, batches of 4,096 pairs, and a fixed 60 epochs without early stopping, so that validation and test data do not influence training.

**Submitted version:** Section 3.6, Eqs. (5)–(7) (p. 6).

**Revised version:** Section 3.5 (p. 4), Eq. (3) (p. 4), Eq. (4) (p. 5); Table 3 (p. 7).

**Comment R3.2.** *"Clarify how positive interactions are selected and how different event types are weighted."*

**Response.** Event weights are now listed: view 0.4, click 0.6, like and bookmark 1.0, complete 1.2, skip $-1.0$, and an explicit rating $r$ maps to $(r-3)/2$. An interaction is positive, and becomes a LightGCN edge, when its weight is positive; repeated events are merged into one edge, and skips and low ratings are excluded. For the Amazon benchmark, ratings of four or five stars are positive.

**Submitted version:** Section 3.5 (p. 6); Section 3.6 (p. 6).

**Revised version:** Section 3.4 (p. 4); Section 3.5 (p. 4); Table 3 (p. 7).

**Comment R3.3.** *"Explain how the EMA profile is initialized and how skips or other negative events update it."*

**Response.** The profile is empty until a user's first non-neutral event, which sets it to the item embedding (or its negation for a negative event); users without events receive no EMA score. Each event with weight $w$ moves the profile by the step $\alpha \min(|w|, 1)$ with $\alpha = 0.25$, toward the item embedding for positive events and toward its negation for skips, and the profile is then renormalized to unit length (Eq. (2) (p. 4)).

**Submitted version:** Section 3.5, Eqs. (3)–(4) (p. 6).

**Revised version:** Section 3.4 (p. 4), Eq. (2) (p. 4).

**Comment R3.4.** *"Provide a precise procedure for constructing KG entities, relations, and user-item paths."*

**Response.** The knowledge-graph subsection now gives a three-step procedure: (i) entities (items typed by content type; creator, genre, and term entities from the catalog text; user and profile nodes), (ii) relations (directed_by, written_by, performed_by, belongs_to_genre, liked, bookmarked, completed, and the profile expansions with examples), and (iii) length-two user–entity–item paths, which are counted by set intersection in the KG and profile scores. Figure 2 was redrawn to match.

**Submitted version:** Section 3.7, Eqs. (9)–(11), Fig. 2 (p. 7).

**Revised version:** Section 3.6 (p. 5); Eq. (5) (p. 5), Eq. (6) (p. 5), Eq. (7) (p. 6); Fig. 2 (p. 5).

**Comment R3.5.** *"Reconcile the stated cross-domain output rate with the scenario results and define the calculation."*

**Response.** The cross-domain rate is now defined as the fraction of scenarios whose top ten results contain at least two content types, and Table A lists the content types of every scenario so that it can be checked directly. All scenarios were re-run through the deployed recommendation API and extended to 17. The rate is 8/17 overall and 8/11 for the scenarios without a domain filter (6/17 and 6/11 if movies and TV shows count as one medium); the six domain-scoped scenarios stay within their declared domain by design.

**Submitted version:** Section 4 and Table 3 (p. 9); abstract, p. 1 (cross-domain output "in 60% of tested scenarios").

**Revised version:** Section 5.3 (p. 9); Table 6 (p. 9).

**Comment R3.6.** *"Include ablation results to show the individual effects of semantic, EMA, graph, and KG/profile signals."*

**Response.** The offline ablation is the experiment described in R2.3 (Table 4 and Table B): the effect of each signal can be read from the 16 combinations on three datasets. In addition, the knowledge-graph and profile signals were ablated in the 17 prototype scenarios, run through the deployed system with each weight set to zero. Without the profile score, mean ProfileAlign@5 falls from 1.00 to 0.96, the mean profile score of the top five from 0.858 to 0.795, and NDCG@10 from 0.943 to 0.916, and on average 0.71 of the top five items change (most strongly for the pop-music user, from 5/5 to 2/5 profile-aligned items). Without the knowledge-graph score, the mean KG score of the top five falls from 0.697 to 0.664 and NDCG@10 from 0.943 to 0.931, with 0.47 of the top five items changed.

**Submitted version:** Section 4 (p. 9).

**Revised version:** Table 4 (p. 8); Section 5.1 (p. 7); Section 5.3 (p. 9).

# Editorial requirements

- **Language:** the manuscript has been proofread and edited throughout.
- **Figures:** both figures are original and were drawn by the authors; no figure is taken from published work.
- **Template:** the manuscript follows the Procedia Computer Science template without modifications (see Comment R1.7).
- **Similarity:** the similarity report of the revised manuscript is uploaded as a supplementary file.

# Additional updates

We also updated the following points in the manuscript:

- the authors of reference [12] (LLMRec), in the References;
- the graph score, now given as the cosine similarity of normalized embeddings, as implemented (Section 3.5 (p. 4));
- the data sources, now matching the current catalog of seven content types and 120,747 items (Section 3.2 (p. 3), Fig. 1 (p. 3));
- the ranking pipeline, which now also describes creator and title matching and the short-term preference bonus, so that Eq. (8) (p. 6) matches the implementation (Section 3.3 (p. 4), Section 3.7 (p. 6)).

We hope that the revised manuscript addresses the reviewers' concerns, and we thank them again for their valuable comments.

Sincerely,
Sanjay A. K., Eswara Raj M., Aditya Vijay, Himaneesh Reddy Jangalapalli, Bagavathi Sivakumar P., Krishna Priya G.

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International Conference on Machine Learning and Data Engineering

# Real-Time Adaptive Cross-Media Recommendation Using Semantic Embeddings, Knowledge Graph Scoring, and Graph Collaborative Learning

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**Abstract**

A user's interests span media, yet most recommenders operate within single-domain silos. This paper presents a real-time adaptive cross-media recommender that maps movies, TV shows, books, and music, extended to health, industrial, and finance products, into one catalog of 120,747 items. Candidates from Sentence-BERT embeddings with Qdrant nearest-neighbor search and from creator and title matching are reranked by a hybrid score combining exponential moving average (EMA) session adaptation, LightGCN collaborative filtering, and knowledge-graph (KG) and profile scoring. On three Amazon Reviews 2023 categories with up to 50,000 users, the hybrid with validation-selected weights is the best of all 16 combinations of its four signals on two categories, improving NDCG@20 over the strongest single signal by 21.6% and 26.9%, and removing any signal lowers it significantly; on the third category, entity overlap alone performs best. A recommendation over the full catalog takes 363 ms at the median on a laptop, and a feedback event updates the user profile in 0.05 ms.

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Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering.

*Keywords:* Recommender systems; Cross-media recommendation; Sentence-BERT; Qdrant; LightGCN; Semantic search; Graph collaborative filtering; Real-time personalization; Knowledge graph

## 1. Introduction

Recommender systems are the main interface through which users discover digital content, yet most deployed systems remain domain-specific: a movie platform recommends films, a music platform recommends tracks, and a book platform recommends books. Human interest does not follow these boundaries. A user interested in science fiction may enjoy a film such as *Interstellar*, a novel about space exploration, a soundtrack with a similar tone, and educational content on astronomy, so a useful recommender should connect these signals across media.

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Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering.

This paper presents a working cross-media recommendation pipeline for movies, TV shows, books, and music, which has since been extended to health, industrial, and finance products. Every item is mapped to one schema and one semantic retrieval layer: Sentence-BERT compares items of different domains through their descriptions [1], Qdrant provides nearest-neighbor retrieval [2], LightGCN learns collaborative preference [3], an exponential moving average (EMA) of item embeddings adapts to each feedback event, and an in-memory knowledge graph (KG) adds entity- and profile-aware scoring, behind a FastAPI backend [4] with MySQL logging.

These capabilities are rarely combined in one multi-domain pipeline: graph models are usually benchmarked on single-domain datasets, semantic systems retrieve by meaning rather than by preference, cross-domain methods typically transfer between two domains instead of normalizing many into one catalog, and real-time personalization is usually separate from graph learning. The contributions of this paper are threefold:

1. An integrated cross-media platform with a unified catalog of seven content types that supports semantic, creator-aware, and title search.
2. A hybrid ranking model that combines semantic retrieval, EMA adaptation, LightGCN graph scoring, and KG/profile scoring, with explainable sub-scores.
3. A quantitative evaluation on three public interaction datasets, comprising single-signal baselines, an ablation over all 16 combinations of the four signals, validation-based weight selection, latency measurements, and scenario tests of the deployed system.

## 2. Related Work and Novelty

Collaborative filtering learns from behavior and content-based recommendation from item metadata, and hybrid approaches combine the two to reduce the cold-start problem [5]. NGCF applies graph convolution to the user–item graph, and LightGCN simplifies it by keeping only neighborhood propagation [6, 3]; here LightGCN is a reranking signal for users with interaction history rather than the candidate generator. Sequential recommenders such as SASRec model the order of user actions within a single domain [7], and Sentence-BERT embeddings [1] let items of different media share one vector space before any interactions exist. KGAT and RippleNet propagate preferences over knowledge-graph relations such as directed-by [8, 9]. Cross-domain recommendation transfers preferences between domains, mostly between two [10], and a recent survey extends the scope to multi-domain transfer [11]; large language models augmented with graph structure have also been used for relevance and explanation [12]. Studies from Amrita Vishwa Vidyapeetham have addressed cold-start prediction, collaborative filtering, sparse data, dual-feedback personalization, and community-aware recommendation [13, 14, 15, 16, 17], but not the integration of heterogeneous media into one pipeline.

Recent work has moved in two directions that are directly relevant here. Knowledge-aware graph recommenders combine LightGCN-style propagation with knowledge graphs: LGKAT adds a personalized knowledge-aware attention sub-network to LightGCN and is evaluated separately on movie, book, music, and restaurant data [18], and LightKG simplifies the graph neural network of knowledge-aware recommenders to reduce training cost on sparse data [19]. Cross-domain recommenders increasingly use large language models as a semantic bridge between domains: LLMCDSR generates pseudo-interactions with an LLM and injects collaborative information into textual item features [20], LLM4CDSR builds unified LLM-based item representations and hierarchical user profiles across domains [21], LeCDSR likewise enhances cross-domain sequential recommendation with an LLM [22], and KLLMRec augments multi-domain recommendation with LLM-derived knowledge [23].

Table 1 contrasts these approaches, based on the components reported in each cited paper. Knowledge-aware graph models learn from one domain at a time, and LLM-based cross-domain models are trained offline for a fixed set of domains around next-item prediction; neither family combines text-based retrieval, collaborative graph scoring, KG/profile scoring, and per-event adaptation in a deployed system. The novelty of this work is therefore not a new neural layer but the integration of these methods over one catalog, with immediate (EMA) and periodic (LightGCN) personalization kept separate; in exchange, its knowledge graph is a lightweight entity-overlap scorer rather than a learned embedding, and its encoder is a compact sentence model rather than an LLM. Because the cited models address different tasks and training protocols, Section 5 does not reimplement them but isolates the contribution of each signal of the proposed pipeline under one protocol.

Table 1. Comparison with representative graph, knowledge-aware, and cross-domain recommenders.

| Method | Domains | Text semantics | Collab. graph | Knowledge graph | Short-term adapt. | Deployed |
|---|---|---|---|---|---|---|
| LightGCN [3] | single | – | yes | – | – | – |
| KGAT [8] | single | – | yes | yes | – | – |
| LGKAT [18] | single | – | yes | yes | – | – |
| LightKG [19] | single | – | yes | yes | – | – |
| LLMCDSR [20] | cross-domain | LLM | – | – | sequence | – |
| LLM4CDSR [21] | cross-domain | LLM | – | – | sequence | – |
| This work | 7 types, one catalog | Sentence-BERT | LightGCN | entity overlap | per-event EMA | yes |

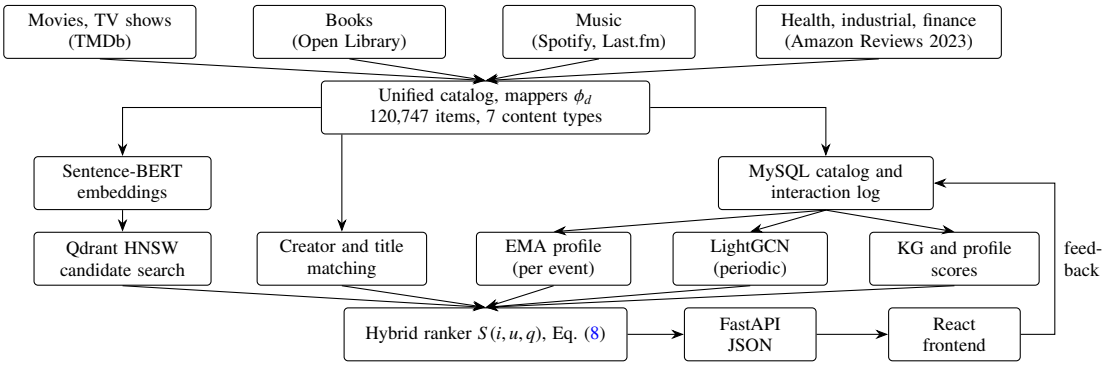



Fig. 1. System architecture of the proposed cross-media recommender.

## 3. Implementation

### 3.1. System overview

In the architecture of Fig. 1, each source is mapped into the unified catalog, embedded, and loaded into Qdrant and MySQL offline, and LightGCN is retrained periodically. At request time, the FastAPI backend validates a query together with an optional user, content-type or domain filter, viewer age, and result count $K$; it matches creators and titles, retrieves Qdrant candidates, reranks them, and returns the final score with every sub-score, which the React frontend displays. Feedback such as likes, skips, and bookmarks is logged through the interaction endpoint and updates the user's EMA profile immediately. The declared viewer age and domain act as a hard eligibility filter before ranking: an undeclared viewer is capped at a teen ceiling, and the health, industrial, and finance catalogs default to adult maturity.

### 3.2. Unified catalog and data sources

Each domain $d \in \mathcal{D}$ provides a mapper $\phi_d$ from a raw row $x$ to the shared item representation $r_i$, and the catalog is

$$C = \bigcup_{d \in \mathcal{D}} \{\phi_d(x) : x \in X_d\}, \quad r_i = (id, type, title, desc, creators, categories, text_i). \tag{1}$$

Adding a domain therefore requires only a new $\phi_d$; embedding, search, KG scoring, the API, and the frontend are unchanged. Movies and TV shows use TMDb metadata and credits [24]; books use the Open Library APIs [25]; music combines a public Spotify songs export with current tracks from the Last.fm API [26]; and health, industrial, and finance products come from the Amazon Reviews 2023 metadata [27], with finance also including accounting and personal-finance titles from Open Library. The deployed catalog holds 120,747 items: 31,133 songs, 23,492 movies,

20,000 health and 20,000 industrial products, 12,402 books, 10,617 TV shows, and 3,103 finance items. Directors and cast, authors, and artists all map to the shared creators field, which therefore serves both person search and the knowledge graph.

### 3.3. Semantic retrieval and creator and title matching

Each item's embedding text (title, categories, description, and creators) and each query are encoded by the all-MiniLM-L6-v2 Sentence-BERT model into 384-dimensional vectors [1]. The users' EMA profiles (Section 3.4) live in the same space, so users, items, and queries are compared directly. With $\mathbf{e}_i$ the embedding of item $i$ and $\mathbf{q}$ that of the query, the semantic score is the cosine similarity $S_{sem}(i,q) = \mathbf{e}_i^\top \mathbf{q}/(\|\mathbf{e}_i\|\,\|\mathbf{q}\|)$. Qdrant performs approximate nearest-neighbor search over this similarity with an HNSW index; it generates candidates rather than the final ranking and lets a new item be recommended from its metadata alone.

Dense similarity does not capture the intent of a name or a short title such as *F1*, so two deterministic stages run first. Creator matching looks for exact, prefix, and substring matches on normalized creator names and places up to five matching items first. Title matching assigns each item one of five tiers on whole normalized words: the query equals the title (tier 0), the title starts with the query (1) or contains it (2), every query token occurs in the title or categories (3), or the query phrase occurs in the description (4). Tiers 0 and 1 rank first and tiers 2–4 are ordered by popularity; lexical matches fill at most $\max(3, \lfloor K/2 \rfloor)$ slots and receive the fixed score $1.30 - 0.06\,t$ for tier $t$, which keeps them above the cosine band. Semantic candidates fill the remaining slots after deduplication.

### 3.4. Interaction logging and EMA adaptation

Interactions are logged in MySQL with a signed weight $w$: view 0.4, click 0.6, like and bookmark 1.0, complete 1.2, and skip $-1.0$; an explicit rating $r \in [1,5]$ maps to $w = (r-3)/2$, and events with $w = 0$ are ignored. Let $\mathbf{u}_t \in \mathbb{R}^{384}$ be the user's profile and $\alpha = 0.25$ the smoothing factor. An event with weight $w$ on item $i$ moves the profile by the step $\eta = \alpha \min(|w|, 1)$ toward $\operatorname{sign}(w)\,\mathbf{e}_i$:

$$\tilde{\mathbf{u}}_{t+1} = (1-\eta)\,\mathbf{u}_t + \eta \operatorname{sign}(w)\,\mathbf{e}_i, \qquad \mathbf{u}_{t+1} = \tilde{\mathbf{u}}_{t+1}/\|\tilde{\mathbf{u}}_{t+1}\|. \tag{2}$$

The profile is empty until the user's first non-neutral event on an embedded item, which sets $\mathbf{u}_1 = \operatorname{sign}(w)\,\mathbf{e}_i$, so users without events receive no EMA score. Skips and low ratings push the profile away, weak signals such as views move it less, and renormalization keeps it on the unit sphere. The EMA score is the cosine similarity $S_{ema}(i,u) = \mathbf{e}_i^\top \mathbf{u}_t/(\|\mathbf{e}_i\|\,\|\mathbf{u}_t\|)$. Because each event updates a single vector, EMA can follow session-level drift without retraining any model.

### 3.5. Graph collaborative filtering with LightGCN

LightGCN [3] is trained on the interaction log. An event with $w > 0$ becomes an unweighted edge of the bipartite graph $\mathcal{G} = (\mathcal{U} \cup \mathcal{I}, \mathcal{E})$, repeated events on the same pair are merged, and skips and low ratings are excluded rather than used as explicit negatives. With $\mathcal{N}_u$ the neighbors of user $u$, layer $\ell$ propagates and the final embedding averages all $L$ layers,

$$\mathbf{e}_u^{(\ell)} = \sum_{i \in \mathcal{N}_u} \frac{\mathbf{e}_i^{(\ell-1)}}{\sqrt{|\mathcal{N}_u|\,|\mathcal{N}_i|}}, \qquad \mathbf{e}_u = \frac{1}{L+1} \sum_{\ell=0}^{L} \mathbf{e}_u^{(\ell)}, \tag{3}$$

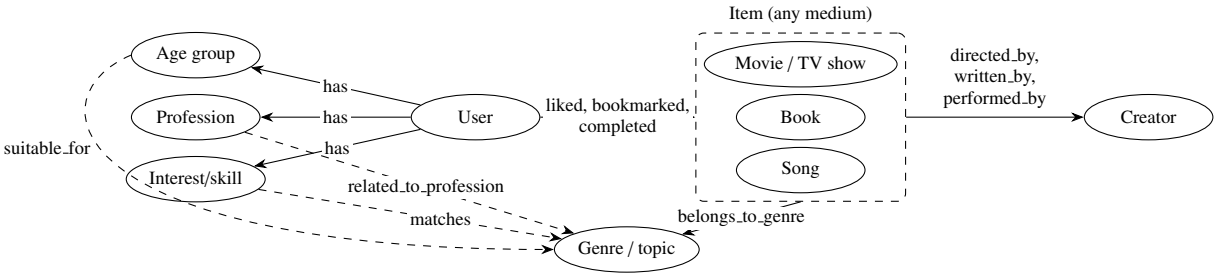



Fig. 2. Knowledge-graph schema; dashed edges are profile expansions into topic terms.

and symmetrically for items. Training minimizes the Bayesian Personalized Ranking (BPR) loss

$$\mathcal{L}_{\text{BPR}} = -\sum_{(u,i,j)\in\mathcal{D}} \ln \sigma(\hat{y}_{ui} - \hat{y}_{uj}) + \lambda\|\Theta\|^2, \tag{4}$$

where $\hat{y}_{ui} = \mathbf{e}_u^\top \mathbf{e}_i$, $\Theta$ are the layer-0 embeddings of the batch, and for every sampled positive pair $(u, i)$ one negative $j$ is drawn uniformly from the catalog and redrawn if $(u, j)$ is observed. At inference time the graph score is the cosine similarity of the $\ell_2$-normalized embeddings, $S_{graph}(i, u) = \mathbf{e}_u^\top \mathbf{e}_i / (\|\mathbf{e}_u\| \, \|\mathbf{e}_i\|)$. A graph score exists only when both user and item are in the training graph; otherwise the candidate is ranked by the other signals. The training settings are given in Section 4.

### 3.6. Knowledge graph and profile-aware scoring

The knowledge graph is built in memory from the catalog, without a separate graph database, in three steps (Fig. 2). *(i) Entities.* Each item becomes a node typed by its content type. After Unicode normalization and case-folding, each creator becomes a Creator node, each category a Genre/topic node, and each alphanumeric token of at least three characters in the title, description, creators, categories, and content type a term entity of the item. Each user becomes a User node, and the declared profile attributes become Profession, Age group, and Interest/skill nodes. *(ii) Relations.* Items link to creators by directed_by, written_by, or performed_by, according to the content type, and to genres by belongs_to_genre; users link to items by liked, bookmarked, or completed when the event weight is positive (these edges also form the LightGCN graph) and to their profile nodes. Each profession and age group expands into a fixed list of related topic terms (related_to_profession and suitable_for; e.g., *data analyst* → data, analytics, business, statistics, technology, science), and an age group may also carry unsuitable terms (e.g., *family* → horror, crime, thriller, war). *(iii) Paths.* A user reaches an item through a length-two path via a shared entity, and because entity namespaces are shared by all domains, the path can end at an item of any medium:

$$P(u, i) = \{u \to e \to i : \; e \in \mathcal{E}_{query} \cup \mathcal{E}_{creator} \cup \mathcal{E}_{genre} \cup \mathcal{E}_{profile}\}. \tag{5}$$

The two scores below count such paths through set intersections, without enumerating them.

The KG score counts paths through query-term entities. With $\mathcal{T}(q)$ and $\mathcal{T}(i)$ the term sets of query $q$ and item $i$,

$$S_{kg}(i, q) = \min\left(1, \frac{|\mathcal{T}(q) \cap \mathcal{T}(i)|}{\max(3, |\mathcal{T}(q)|)}\right), \tag{6}$$

where the floor of three prevents a one- or two-word query from reaching the maximum through a single shared term. The profile score rewards items that match the user's declared attributes and penalizes unsuitable ones:

$$S_{profile}(i,u) = \sum_{t \in \mathcal{P}_+(u)} \delta(t,i) w_+ - \sum_{t \in \mathcal{P}_-(u)} \delta(t,i) w_- + \mathbf{1}[\text{pref}(u) = \text{type}(i)] w_c, \tag{7}$$

where $\mathcal{P}_+(u)$ holds the user's interests, skills, and the terms expanded from the profession and age group, $\mathcal{P}_-(u)$ the unsuitable terms, $\delta(t,i) = 1$ if term $t$ occurs in the text of item $i$, $\text{pref}(u)$ the preferred content types, $w_+ = 0.16$, $w_- = 0.25$, and $w_c = 0.15$; the score is clipped to $[-1, 1]$. The profile uses only declared, consent-based attributes, not private social-media data.

### *3.7. Hybrid ranking*

Candidates are sorted by the hybrid score

$$S(i,u,q) = S_{sem}(i,q) + B_{pref}(i,u) + \lambda_g S_{graph}(i,u) + \lambda_e S_{ema}(i,u) + \lambda_k S_{kg}(i,q) + \lambda_p S_{profile}(i,u), \tag{8}$$

where $S_{sem}$ is replaced by the fixed lexical score for creator and title matches. $B_{pref}$ is a short-term preference bonus from the user's 500 most recent events: with $c_u(g)$ and $c_u(\tau)$ the summed event weights of category $g$ and content type $\tau$,

$$B_{pref}(i,u) = 0.035 \sum_{g \in \text{cat}(i)} \min(c_u(g), 3) + 0.02 \min(c_u(\text{type}(i)), 3) - 0.15\, \mathbf{1}[i \in \mathcal{I}_u^+] - 0.35\, \mathbf{1}[i \in \mathcal{I}_u^-], \tag{9}$$

where $\mathcal{I}_u^+$ and $\mathcal{I}_u^-$ are the items the user has already rated positively and negatively, which are demoted so that new items are favored. For known users, $5K$ candidates are retrieved before reranking, so that personalization can promote items from outside the initial top $K$. The deployed system uses the hand-set weights $\lambda_g = 0.20$, $\lambda_e = 0.15$, $\lambda_k = 0.08$, and $\lambda_p = 0.10$; Section 5 compares them with weights selected on a validation split.

## 4. Experimental Setup

An offline benchmark on public interaction data measures every signal of Eq. (8) under identical conditions; scenario tests of the deployed system cover the query- and profile-dependent behavior.

*Datasets.* The benchmark uses three categories of the Amazon Reviews 2023 corpus [27]: Movies & TV, Health & Household, and Industrial & Scientific. Ratings of four or five stars are positive interactions; other ratings are discarded. The two large categories are filtered to a 10-core graph and a random sample of 50,000 users is drawn, after which items with fewer than five remaining interactions are removed; Industrial & Scientific does not survive a 10-core filter and uses the 5-core graph without sampling. Item text (title, category path, description, and store) comes from the corpus metadata. Table 2 lists the graphs, whose densities are about 0.04%; metadata covers every Health & Household and Industrial & Scientific item but only 67.3% of the Movies & TV items, and items without text receive no semantic, EMA, or KG signal.

*Split, candidates, and metrics.* Each user's positives are ordered by timestamp; the last two form the test set, the one before them the validation set, and the rest the training set, on which alone LightGCN is trained. Every method ranks the full training catalog minus the user's history, so evaluation involves no negative sampling. Recall@$K$, NDCG@$K$, MAP@$K$, and MRR@$K$ are averaged over test users for $K \in \{10, 20\}$ (HitRate@$K$ and Precision@$K$ are computed as well), and significance is assessed with a paired Wilcoxon signed-rank test on per-user NDCG@20.

*Offline signals.* Public logs contain neither typed queries nor declared profiles, so each signal uses its closest offline counterpart: $S_{sem}$ takes the user's most recent history item as the query, which corresponds to the system's similar-item endpoint; $S_{ema}$ replays Eq. (2) over the history in time order; $S_{graph}$ is the LightGCN cosine score; and $S_{kg}$ applies Eq. (6) to the entities of the user's history (category-path nodes, store, and main category), normalized by the candidate's entity count because a history holds far more entities than a query. Entities occurring on a single item or on more than half of the items are dropped. The declared-profile score has no offline counterpart and is covered by the scenarios, and $B_{pref}$ is omitted because its category term overlaps with the offline KG signal.

*Baselines, combinations, and weight selection.* Most-popular ranking uses neither content nor personalization, and each single-signal baseline keeps one source of evidence: semantic similarity to the latest item (content only, usable without interactions), EMA similarity to the session profile (recent taste only), the LightGCN score (collaborative behavior only), and KG entity overlap (shared categories and brands only). The ablation evaluates all $2^4 = 16$ combinations of the four signals, from the empty one (popularity alone) to all four, plus the hybrid with the hand-set weights of Section 3.7. Each combination has its own weights, chosen by grid search on 10,000 validation users to maximize NDCG@20 (the first signal has weight 1 and every other signal a weight in $\{0.05, 0.1, 0.2, 0.4, 0.8, 1.6, 3.2\}$), so that no combination is handicapped by weights chosen for another; the test set is used once.

*Training settings.* Table 3 lists the settings. LightGCN is trained for a fixed 60 epochs of $\lceil |\mathcal{E}|/B \rceil$ mini-batches without early stopping, so that neither validation nor test data influence training. All runs use seed 42 on one NVIDIA T4 GPU, on which a training run takes 17.0, 13.2, and 1.2 minutes on the three graphs.

Table 2. Benchmark graphs after filtering.

| Category | Users | Items | Training positives | Test users | Items with text |
|---|---|---|---|---|---|
| Movies & TV | 50,000 | 52,213 | 814,346 | 49,994 | 35,153 (67.3%) |
| Health & Household | 50,000 | 42,621 | 709,212 | 50,000 | 42,621 (100.0%) |
| Industrial & Scientific | 33,959 | 18,804 | 170,834 | 33,926 | 18,804 (100.0%) |

Table 3. LightGCN and EMA settings.

| Setting | Value | Setting | Value |
|---|---|---|---|
| Embedding size / layers | 64 / 3, averaged (Eq. (3)) | Negative sampling | 1 uniform per positive, resampled if observed |
| Initialization | $\mathcal{N}(0, 0.1^2)$ | Positive interactions | $w > 0$ (Amazon: rating $\geq 4$), merged |
| Optimizer / lr / $L_2$ $\lambda$ | Adam / $10^{-3}$ / $10^{-4}$ | EMA $\alpha$ / step | 0.25 / $\eta = \alpha \min(\lvert w \rvert, 1)$ |
| Batch / epochs | 4096 / 60, no early stopping | Text encoder | all-MiniLM-L6-v2, 384-d |

## 5. Results

### 5.1. Baselines and ablation

Table 4 reports the 16 combinations of the four signals at $K = 20$ (S: semantic, E: EMA, G: LightGCN, K: knowledge graph): the empty combination is most-popular ranking, the four single-signal rows are the baselines, S+E+G+K is the proposed hybrid, and the last row is the hybrid with the hand-set weights. Absolute values are low, as usual for full-catalog ranking on sparse purchase data, so methods are compared under identical conditions. Every combination differs significantly from S+E+G+K in NDCG@20 ($p < 0.01$, Wilcoxon) except the one marked °, and the results at $K = 10$ and for HitRate and Precision lead to the same conclusions.

On Movies & TV and Industrial & Scientific, the four-signal hybrid is the best of the 16 combinations on all four measures at both cut-offs. It improves NDCG@20 over the strongest single signal, semantic similarity, by 21.6% and 26.9% (most-popular: factors of 2.3 and 1.9) and HitRate@20 from 0.051 to 0.063 and from 0.068 to 0.089. The signals are complementary: LightGCN alone is weaker than semantic retrieval on every dataset, yet semantic plus LightGCN is the best pair everywhere, and removing either from the hybrid lowers NDCG@20 by 8.6–22.0% (semantic) and 13.4–21.9% (LightGCN) on the three datasets. Removing the KG score costs 1.5–3.0% on every

Table 4. Offline ranking results at $K = 20$ for the 16 signal combinations (R: Recall, N: NDCG; best value in bold).

| Signals | Movies & TV | | | | Health & Household | | | | Industrial & Scientific | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | R@20 | N@20 | MAP@20 | MRR@20 | R@20 | N@20 | MAP@20 | MRR@20 | R@20 | N@20 | MAP@20 | MRR@20 |
| None (popularity) | 0.0197 | 0.0097 | 0.0048 | 0.0089 | 0.0237 | 0.0115 | 0.0056 | 0.0106 | 0.0282 | 0.0147 | 0.0077 | 0.0150 |
| S | 0.0307 | 0.0187 | 0.0118 | 0.0196 | 0.0166 | 0.0096 | 0.0054 | 0.0104 | 0.0379 | 0.0223 | 0.0134 | 0.0238 |
| E | 0.0233 | 0.0128 | 0.0074 | 0.0126 | 0.0104 | 0.0052 | 0.0026 | 0.0050 | 0.0254 | 0.0135 | 0.0074 | 0.0131 |
| G | 0.0257 | 0.0136 | 0.0075 | 0.0134 | 0.0116 | 0.0054 | 0.0025 | 0.0047 | 0.0242 | 0.0130 | 0.0072 | 0.0130 |
| K | 0.0093 | 0.0045 | 0.0021 | 0.0041 | **0.0253** | **0.0144** | **0.0081** | **0.0157** | 0.0364 | 0.0210 | 0.0122 | 0.0224 |
| S+E | 0.0309 | 0.0188 | 0.0118 | 0.0197 | 0.0169 | 0.0097 | 0.0055 | 0.0105 | 0.0380 | 0.0225 | 0.0135 | 0.0240 |
| S+G | 0.0359 | 0.0222 | 0.0143 | 0.0238 | 0.0212 | 0.0120 | 0.0068 | 0.0129 | 0.0468 | 0.0267 | 0.0156 | 0.0278 |
| S+K | 0.0313 | 0.0193 | 0.0123 | 0.0205 | 0.0170 | 0.0097 | 0.0055 | 0.0105 | 0.0387 | 0.0226 | 0.0135 | 0.0240 |
| E+G | 0.0307 | 0.0174 | 0.0104 | 0.0177 | 0.0187 | 0.0092 | 0.0046 | 0.0086 | 0.0377 | 0.0205 | 0.0116 | 0.0205 |
| E+K | 0.0247 | 0.0141 | 0.0085 | 0.0143 | 0.0144 | 0.0072 | 0.0036 | 0.0069 | 0.0314 | 0.0162 | 0.0087 | 0.0154 |
| G+K | 0.0265 | 0.0141 | 0.0078 | 0.0139 | 0.0200 | 0.0097 | 0.0047 | 0.0091 | 0.0386 | 0.0207 | 0.0115 | 0.0205 |
| S+E+G | 0.0362 | 0.0224 | 0.0144 | 0.0240 | 0.0215 | 0.0122 | 0.0069 | 0.0131 | 0.0490 | 0.0277 | 0.0162 | 0.0286 |
| S+E+K | 0.0318 | 0.0197 | 0.0126 | 0.0209 | 0.0171 | 0.0098 | 0.0056 | 0.0107 | 0.0389 | 0.0227 | 0.0136 | 0.0241 |
| S+G+K | 0.0366 | 0.0226 | 0.0145 | 0.0241 | 0.0224 | 0.0126° | 0.0071 | 0.0135 | 0.0484 | 0.0275 | 0.0161 | 0.0286 |
| E+G+K | 0.0314 | 0.0177 | 0.0106 | 0.0179 | 0.0232 | 0.0115 | 0.0058 | 0.0109 | 0.0410 | 0.0225 | 0.0128 | 0.0227 |
| S+E+G+K | **0.0368** | **0.0227** | **0.0146** | **0.0243** | 0.0223 | 0.0126 | 0.0071 | 0.0135 | **0.0495** | **0.0283** | **0.0167** | **0.0295** |
| Hand-set weights | 0.0344 | 0.0216 | 0.0140 | 0.0230 | 0.0208 | 0.0122 | 0.0070 | 0.0133 | 0.0455 | 0.0266 | 0.0159 | 0.0282 |

dataset ($p < 10^{-6}$), and removing EMA 0.7% and 2.7% on Movies & TV and Industrial & Scientific ($p < 0.01$), with no significant effect on Health & Household.

Health & Household behaves differently. Semantic similarity, EMA, and LightGCN on their own all rank below most-popular, whereas KG overlap alone, which here matches category paths and brands, is the best combination on every measure and exceeds the hybrid by 14.4% in NDCG@20; the hybrid ties S+G+K and beats the other 13 combinations. Repeat purchases within a brand or category, which entity overlap captures directly, dominate this domain. The validation split, with one held-out item per user, nevertheless selected a small KG weight, so weights should be tuned per domain on more validation data.

The selected four-signal weights are stable: relative to the semantic score, $\lambda_g = 0.8$, $\lambda_e = 0.2$–$0.4$, and $\lambda_k = 0.05$–$0.1$ on all three datasets. The hand-set weights thus under-weight the graph signal by a factor of four, and the tuned weights improve NDCG@20 over them by 5.4%, 3.6%, and 6.5% ($p < 10^{-4}$), although the two tie on Health & Household at $K = 10$.

### 5.2. Latency

Latency was measured on a laptop (AMD Ryzen 5 5600H, 16 GB RAM, NVIDIA RTX 3050 Laptop GPU for query encoding) running MySQL 8.0 and a Qdrant 1.12.6 server with an HNSW index ($m = 16$, ef_construct = 100) over the full 120,747-item catalog. After 30 warm-up requests, 700 requests with $K = 10$ (16 queries, each issued anonymously and for six demo users) were sent through FastAPI's in-process test client, so validation, ranking, and serialization are included and only the network hop is excluded (Table 5).

Over all 700 requests, the median latency is 363 ms (mean 371 ms, p95 455 ms, p99 507 ms). Personalized requests are 28 ms slower at the median than anonymous ones: the MySQL preference bonus (mean 6.4 ms), LightGCN (0.07 ms), and EMA stages add about 15 ms, and the rest comes from retrieving $5K$ instead of $K$ candidates. An EMA update takes 0.05 ms, so a like or skip is reflected in the very next request without retraining. The dominant costs are the lexical title scan (191 ms), which scans the catalog with pandas on every request, and the Qdrant query (83 ms), which returns up to $5K$ candidates with their payloads. Because the CPU-bound scan is serialized by the Python interpreter lock, one API process sustains 2.7 requests per second, and 3.3 with eight concurrent workers; an inverted title index and several worker processes are the main routes to higher throughput.

Table 5. Latency of *POST /recommend* by request type and stage, in milliseconds.

| Request or stage | Mean | p50 | p95 | p99 |
|---|---|---|---|---|
| End-to-end, anonymous ($n$ = 100) | 345.9 | 342.1 | 426.6 | 488.8 |
| End-to-end, personalized ($n$ = 600) | 375.5 | 369.8 | 458.8 | 508.4 |
| Creator matching | 46.2 | 44.4 | 63.1 | 74.1 |
| Lexical title/keyword matching | 198.9 | 190.8 | 265.7 | 307.2 |
| Query encoding (Sentence-BERT) | 12.6 | 12.0 | 14.0 | 23.4 |
| Qdrant HNSW search with payloads | 85.2 | 82.6 | 106.3 | 126.4 |
| EMA rerank | 6.5 | 7.3 | 9.3 | 10.6 |
| KG/profile rerank | 13.1 | 13.1 | 22.3 | 28.2 |

### *5.3. Prototype scenarios*

Seventeen scenarios were issued through the deployed endpoint with $K = 10$, each as a seeded demo user with the viewer age implied by its profile (20, 30, or 8 years; vertical queries also declare their domain): five person queries, six cross-media intents, and six vertical queries (health, industrial, and finance, two each). Table 6 gives means by kind of scenario, and every scenario is listed in the supplementary material. PersonHit@5 (PH) counts the top-5 results that credit the queried person, ProfileAlign@5 (PA) those with a positive profile score (Eq. (7)), and KG and Prof. are the mean $S_{kg}$ and $S_{profile}$ of the top five. Typed queries have no human labels, so the rank metrics at 10 use rule-based relevance judgments: an item is relevant if the queried person is among its creators, or if its title, categories, creators, or description contain at least two of the scenario's intent terms (within the declared vertical, if any). The rule reads catalog metadata only, never a ranker score, and recall divides by all relevant items (15 to 6,558), so it is at most $10/|\text{relevant}|$. The cross-domain rate is the fraction of scenarios whose top ten contains at least two content types.

Table 6. Prototype scenarios run through the deployed recommendation API, averaged by kind.

| Scenarios | $n$ | PH@5 | PA@5 | KG | Prof. | P@10 | R@10 | N@10 | MAP@10 | MRR@10 | Cross-domain |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Person queries | 5 | 0.88 | 1.00 | 0.600 | 0.717 | 0.840 | 0.329 | 0.821 | 0.729 | 0.800 | 4/5 |
| Cross-media intents | 6 | – | 1.00 | 0.686 | 0.969 | 0.983 | 0.006 | 0.988 | 0.978 | 1.000 | 4/6 |
| Vertical queries | 6 | – | 1.00 | 0.788 | 0.864 | 1.000 | 0.009 | 1.000 | 1.000 | 1.000 | 0/6 |
| All scenarios | 17 | 0.88 | 1.00 | 0.697 | 0.858 | 0.947 | 0.102 | 0.943 | 0.913 | 0.941 | 8/17 |

All five person queries place four or five of the person's items in the top five; the misses carry the name in the title without listing the person as a creator, such as *James Cameron's Story of Science Fiction*. Eight of the eleven scenarios without a domain filter return at least two content types (6 of 11 if movies and TV shows count as one medium), the family profile receives only age-appropriate films, and the vertical scenarios stay in their declared domain by design, so the cross-domain rate is 8/17 overall. Eleven of the twelve topical queries score 1.00 on every rank measure, so they mainly confirm that results match the intent terms; the person queries discriminate (mean P@10 0.84, NDCG@10 0.82, MAP@10 0.73, MRR@10 0.80). EMA scores were available for all 170 returned items but a LightGCN score for only seven, because the deployed LightGCN model is trained on the demo log (13 users, 347 items); its contribution is measured by the benchmark.

The KG and profile signals have no offline counterpart, so they were ablated in the scenarios by setting their weights to zero. Without the profile score, mean ProfileAlign@5 falls from 1.00 to 0.96, the mean profile score of the top five from 0.858 to 0.795, NDCG@10 from 0.943 to 0.916, and 0.71 of the top five items change; the effect is largest for the pop-music user (ProfileAlign@5 from 5/5 to 2/5 as a concert film replaces an album) and the broad space-adventure query (profile score 0.896 to 0.560). Without the KG score, the mean KG score falls from 0.697 to 0.664, NDCG@10 from 0.943 to 0.931, and 0.47 of the top five items change. Person matching is unchanged, the cross-domain rate rises from 8/17 to 9/17 in both variants (one scenario gains a TV show), and no variant returns an unsuitable item, because the eligibility filter of Section 3 removes such items before ranking.

## 6. Conclusion and Future Work

This paper presented a real-time adaptive cross-media recommender that ranks seven content types from one catalog by combining semantic, creator and title, EMA, LightGCN, and KG and profile signals. On three Amazon Reviews 2023 categories, the hybrid with validation-selected weights was the best of all 16 signal combinations on two of them, the semantic and LightGCN signals contributed most, and weight selection improved on the hand-set weights; a request takes 363 ms at the median and a feedback update 0.05 ms. Its limitations are the Health & Household result, where entity overlap alone was best, the demo-only coverage of the deployed LightGCN model, and a throughput of about three requests per second; future work will address them with per-domain weight tuning and an inverted title index, and will add further domains and user studies.

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